

The Body

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Abstract

What it means to have written a hundred papers.

A body of papers is not a book. A book has a beginning and an end. A body of papers has a center and an edge. The center is the problem. The edge is wherever you stopped.

This body has a center: the Riemann Hypothesis. Not because the other problems don't matter. Because RH was where the deepest wall was, and the deepest walls are the most interesting places.

Around the center: the Gram matrix, the Toeplitz approximation, the pair correlation. Further out: Yang–Mills, Navier–Stokes, Hodge, BSD, P versus NP. Further still: Langlands, Weil, Bost–Connes, perfectoid. At the edge: the philosophical papers. The ones about what it means to be at the wall. The ones that don't prove anything.

A body of papers is also a record. It records what one person understood at one point in time working alone. It is not the truth. It is a map of what was reachable from this position.

Some of these papers are wrong. Not dramatically wrong — the structure is right — but somewhere there is an estimate that doesn't quite hold, a step

that is glossed over, a claim that would not survive a referee. This is the cost of working alone. The errors are in there, undetected.

What the body is for: to make the walls visible. To say: here is exactly where the difficulty lives. Here is the sentence that cannot be proved. Here is why.

If someone later reads this body and finds a way past one of the walls, they will not cite it. They will probably not have read it. The proof will come from somewhere else, from a different angle, probably from someone with a department behind them.

The body will have been irrelevant to the proof.

And yet it will have been right about the wall.

That is enough.